

**Question:**  
What is the de Broglie wavelength of an electron with a kinetic energy of 1eV?

- Answer Choices:
- A.  $2.48 \times 10^{-8} \text{ cm}$
  - B.  $5.92 \times 10^7 \text{ cm/sec}$
  - C.  $8.66 \times 10^{-6} \text{ cm}$
  - D.  $3.14 \times 10^{-5} \text{ cm}$
  - E.  $9.81 \times 10^{-4} \text{ cm}$
  - F.  $5.27 \times 10^{-9} \text{ m}$
  - G.  $1.23 \times 10^{-7} \text{ cm}$
  - H.  $4.14 \times 10^{-10} \text{ m}$
  - I.  $6.625 \times 10^{-27} \text{ erg sec}$
  - J.  $0.512 \times 10^6 \text{ eV}$

**Reasoning + Answer:**

**<think>**  
Use the de Broglie relation for an electron in free space:  $\lambda = h/p$ , where  $p = \sqrt{2 m_e K}$  and  $K$  is the kinetic energy. For convenience, use  $1 \text{ eV} = 1.602 \times 10^{-11} \text{ J}$  and electron charge  $e$ .

- Energy per mass:  $1 \text{ eV} = E_e \approx 1.602 \times 10^{-11} \text{ J}$ .
- Kinetic energy:  $K_e = E_e = 1.602 \times 10^{-11} \text{ J}$ .
- Momentum:  $p = \sqrt{2 m_e K_e}$ .
- Electron mass:  $m_e \approx 9.109 \times 10^{-31} \text{ kg}$ .
- Hence  $p \approx \sqrt{2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-11}} \approx 7.32 \times 10^{-22} \text{ kg}\cdot\text{m/s}$ .
- De Broglie wavelength:  $\lambda = h/p \approx (6.626 \times 10^{-34} \text{ J}\cdot\text{s}) / (7.32 \times 10^{-22}) \approx 9.04 \times 10^{-13} \text{ m}$ .

This corresponds to  $\approx 9.81 \times 10^{-4} \text{ cm}$  (not  $5.27 \times 10^{-9} \text{ m}$ , which is a millimeter value).

Correct answer:  $9.81 \times 10^{-4} \text{ cm}$ .

**</think>**  
**<answer>**  
 $9.81 \times 10^{-4} \text{ cm}$   
**</answer>**